

Cost Analysis

2026-06-29

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1 Assumptions

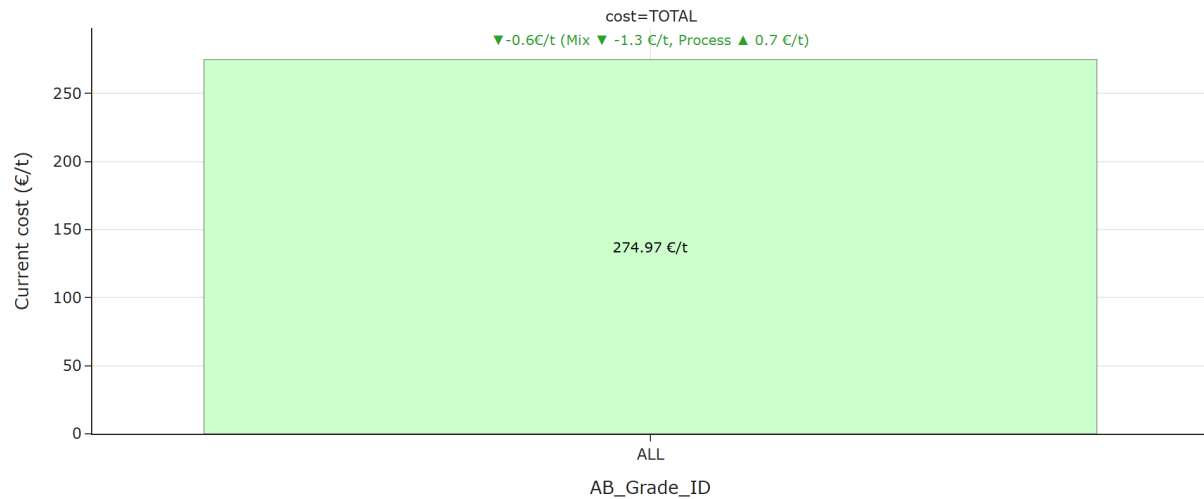
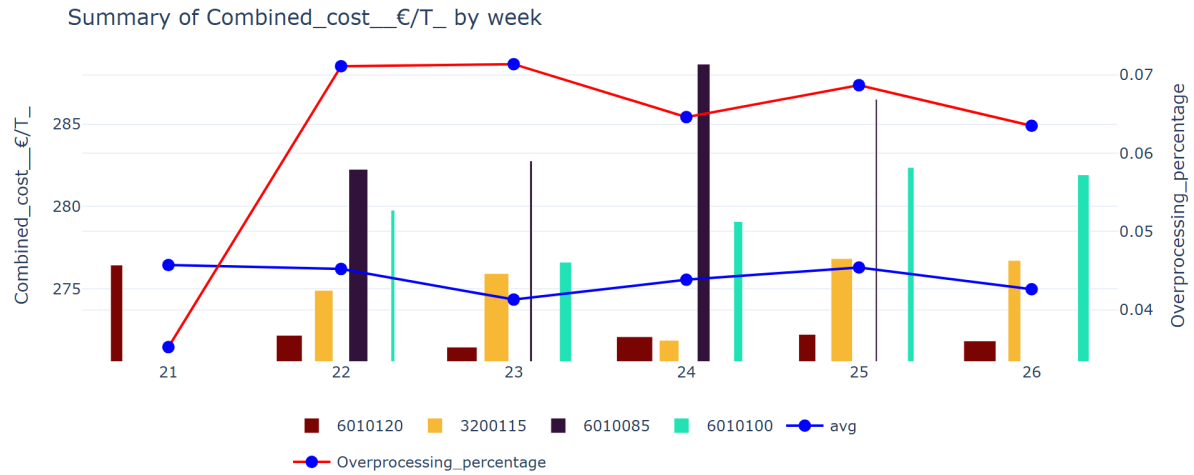
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

- **Period Analyzed:** May 24, 2026 – June 29, 2026
- **Cost Components Included:** Fibre, steam, electricity, starch, and chemicals.
- **Grades Included:** Only grades 6010085, 6010100, 601020 and 3200115 are included in this week’s analysis.
- **Data Selection Criteria:** Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used:**
 - **Current:** Refers to the last week of the analyzed period.
 - **Historic:** Represents the average cost over the monthly interval previous to the “Current” period. Used as baseline.
 - **Best:** Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

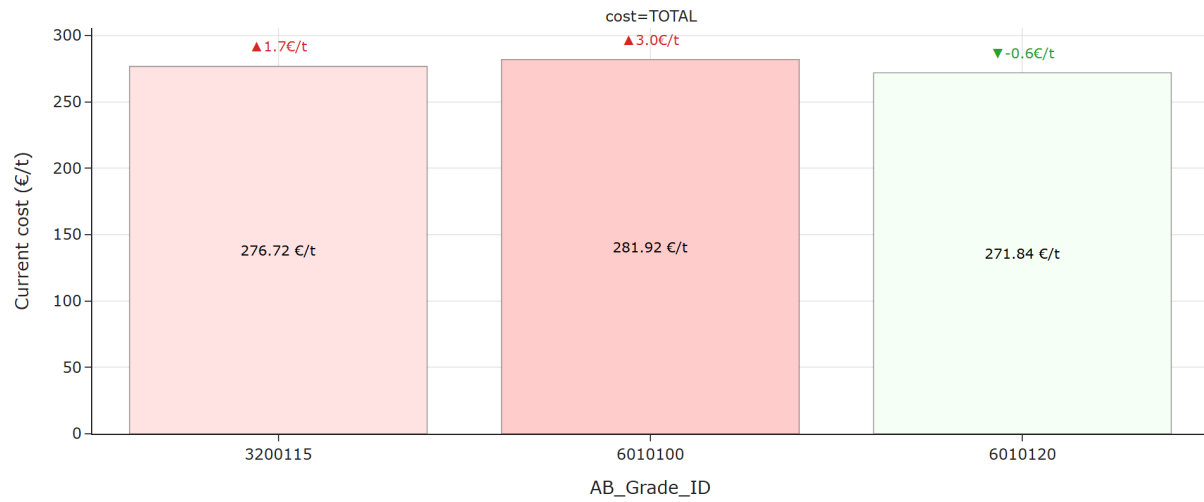
The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.



For all considered grades, TOTAL cost saw a decrease of 0.60 €/t, resulting in 274.97 €/t.

Breaking down the overall cost change, -1.28 €/t was driven by a cheaper mix of grades, and +0.68 €/t was due to a deterioration in the process.

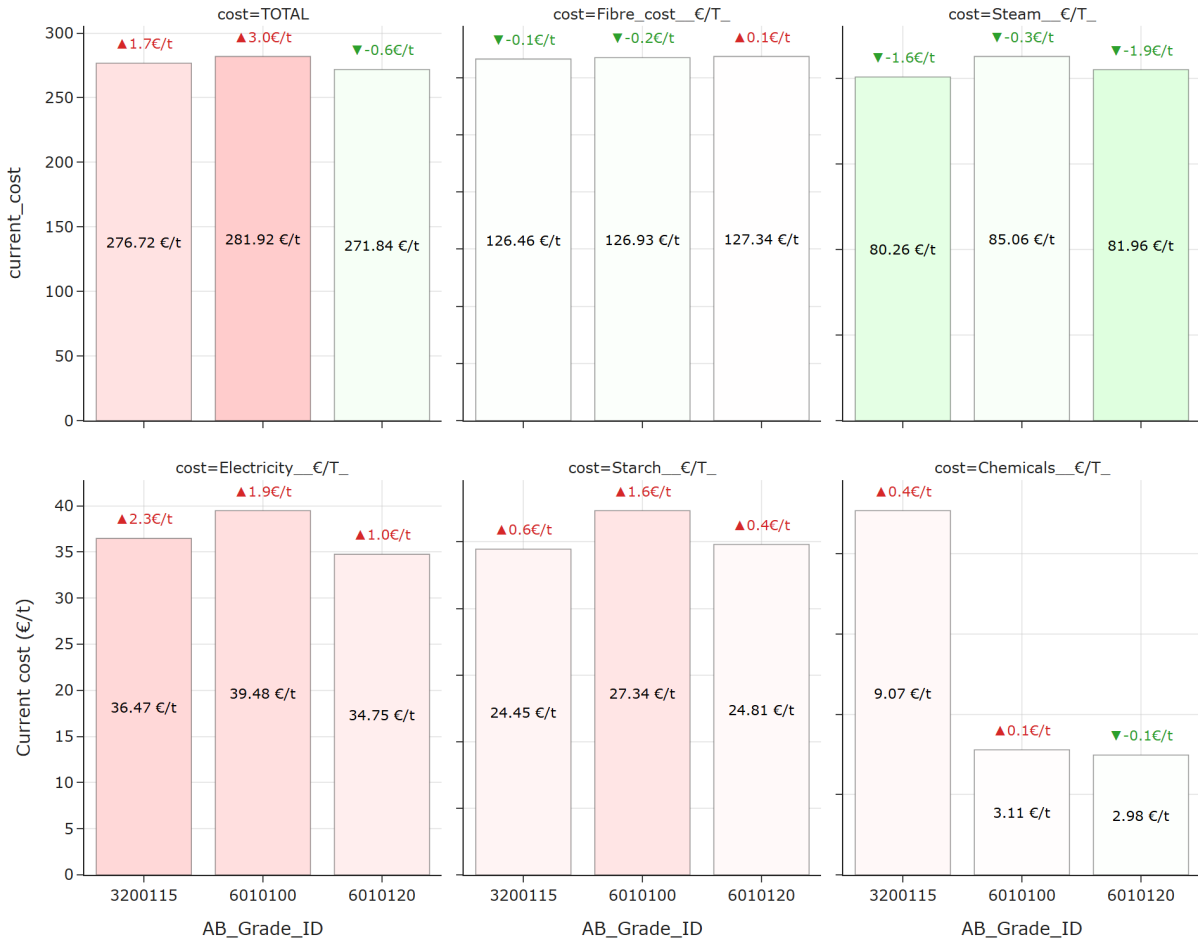


For grade 3200115, TOTAL cost saw an increase of 1.73 €/t, resulting in 276.72 €/t.

For grade 6010100, TOTAL cost saw an increase of 3.04 €/t, resulting in 281.92 €/t.

For grade 6010120, TOTAL cost saw a decrease of 0.56 €/t, resulting in 271.84 €/t.

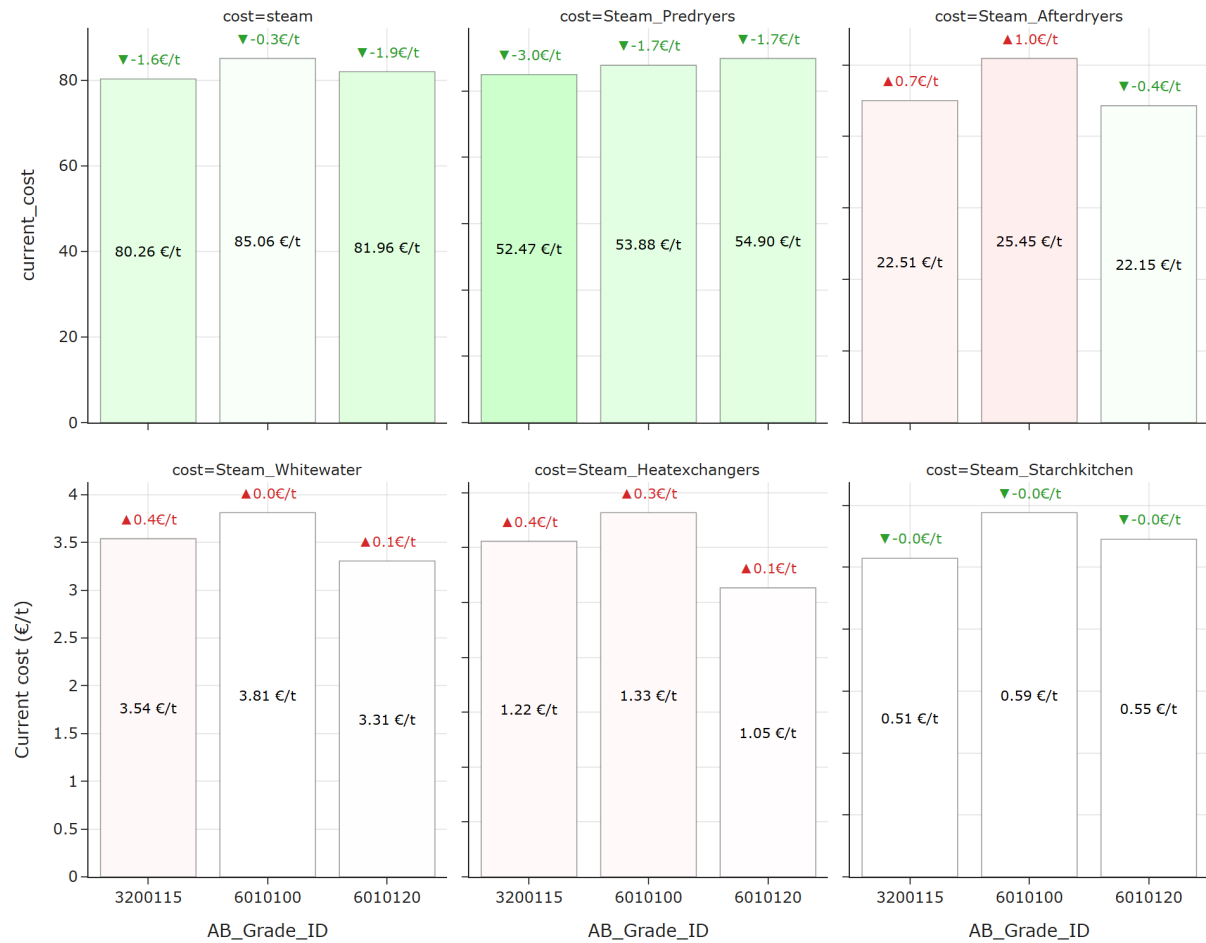
Looking across individual grades, the strongest cost increase was observed for grade 6010100 (+3.04 €/t) and the strongest cost decrease was observed for grade 6010120 (-0.56 €/t). These movements stand out and may require further investigation.



Average change by component:

- **Electricity_€/T** increased on average (+1.77 €/t).
- **Steam_€/T** decreased on average (-1.28 €/t).
- **Starch_€/T** increased on average (+0.85 €/t).
- **Chemicals_€/T** increased on average (+0.14 €/t).
- **Fibre_cost_€/T** decreased on average (-0.08 €/t).

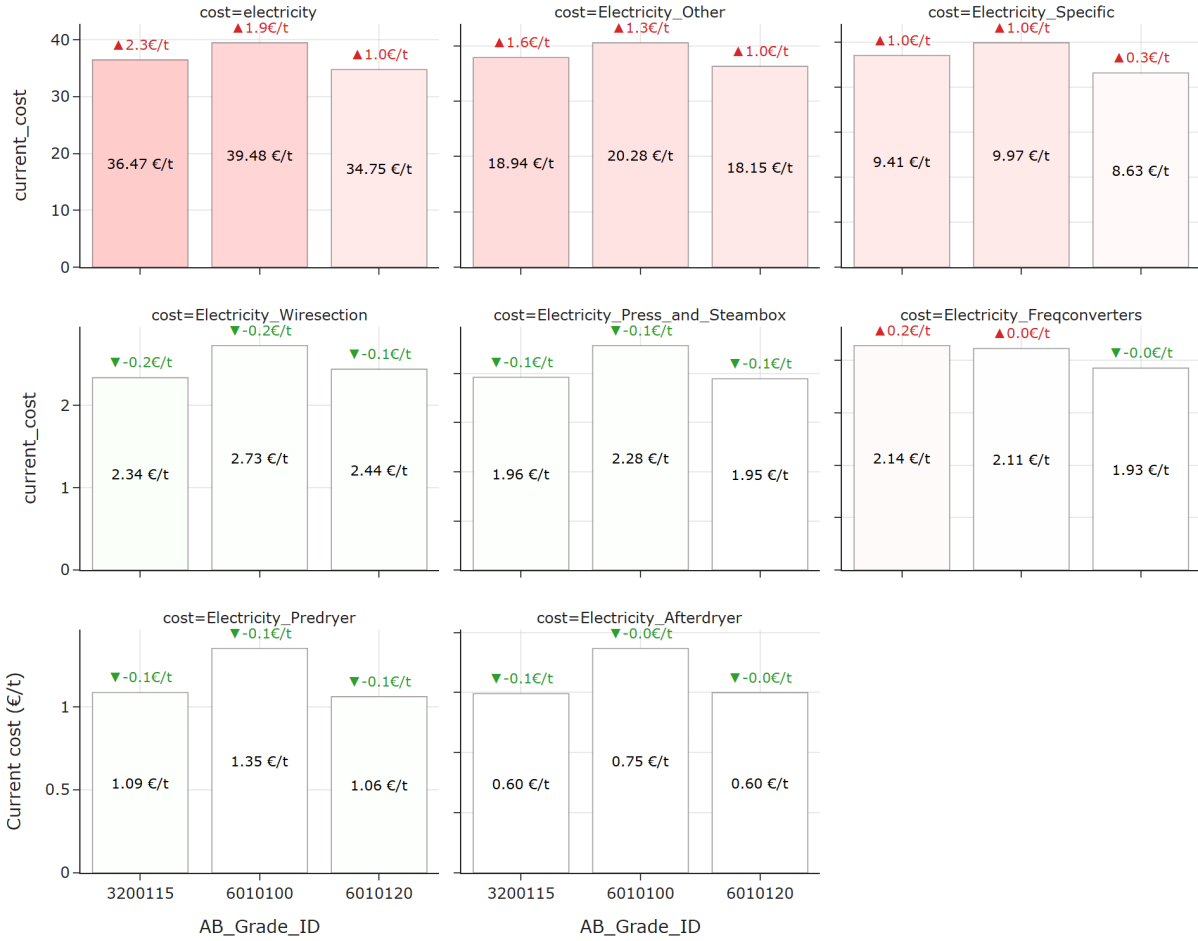
A more detailed breakdown by component shows that the largest average increase comes from **Electricity_€/T** (+1.77 €/t on average), with the highest increase in grade 3200115 (+2.35 €/t) and the largest average decrease comes from **Steam_€/T** (-1.28 €/t on average), with the biggest drop in grade 6010120 (-1.91 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam_Predryers** decreased on average (-2.16 €/t).
- **Steam_Afterdryers** increased on average (+0.43 €/t).
- **Steam_Heatexchangers** increased on average (+0.28 €/t).
- **Steam_Whitewater** increased on average (+0.17 €/t).
- **Steam_Starchkitchen** decreased on average (-0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Afterdryers** (+0.43 €/t on average), with the highest increase in grade 6010100 (+1.00 €/t) and the largest average decrease comes from **Steam_Predryers** (-2.16 €/t on average), with the biggest drop in grade 3200115 (-3.05 €/t). These components stand out and may require further investigation.

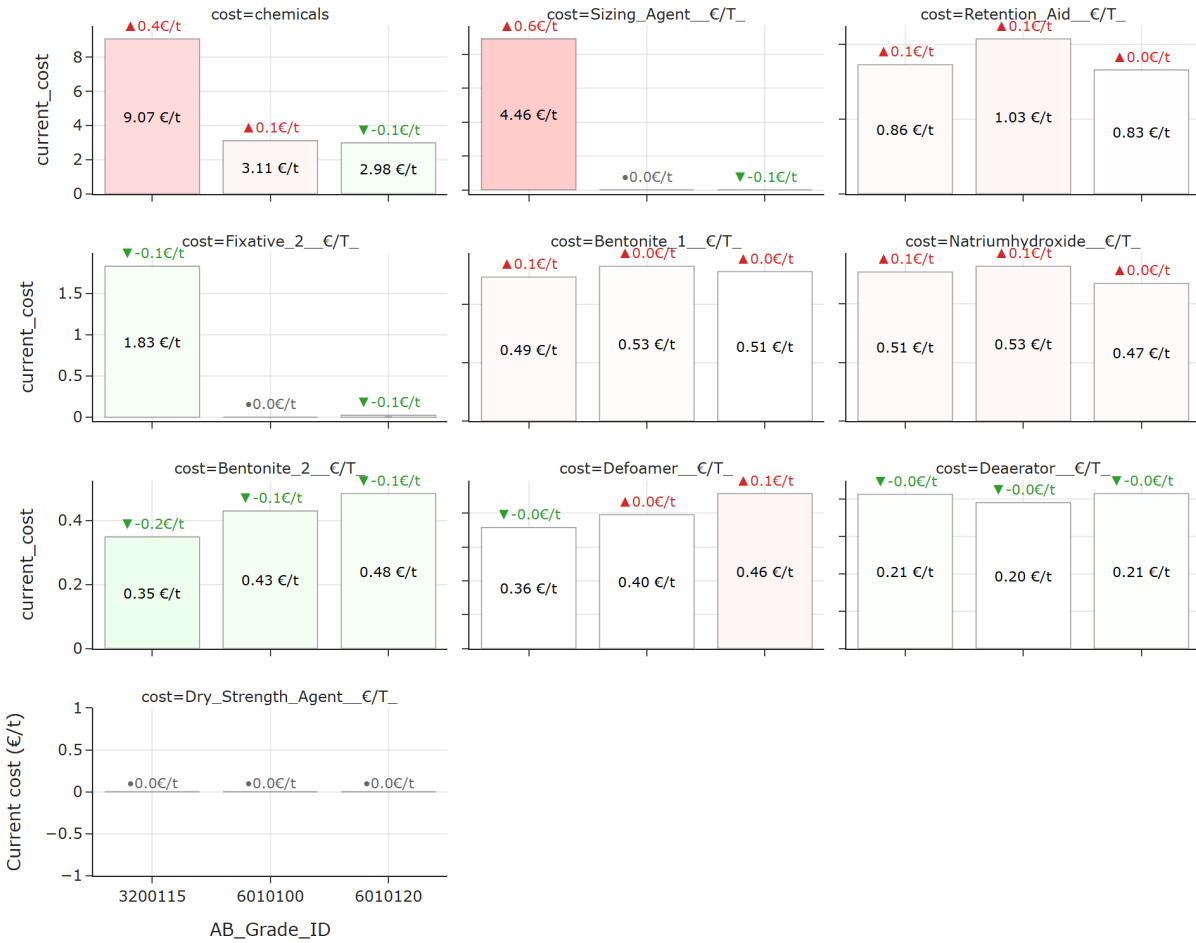


Average change by electricity:

- **Electricity_Other** increased on average (+1.31 €/t).
- **Electricity_Specific** increased on average (+0.76 €/t).
- **Electricity_Wiresection** decreased on average (-0.17 €/t).
- **Electricity_Press_and_Steambox** decreased on average (-0.11 €/t).
- **Electricity_Freqconverters** increased on average (+0.09 €/t).
- **Electricity_Predryer** decreased on average (-0.08 €/t).
- **Electricity_Afterdryer** decreased on average (-0.05 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Other** (+1.31 €/t on average), with the highest increase in grade 3200115 (+1.63 €/t) and the largest average decrease comes from **Electricity_Wiresection** (-0.17 €/t on average),

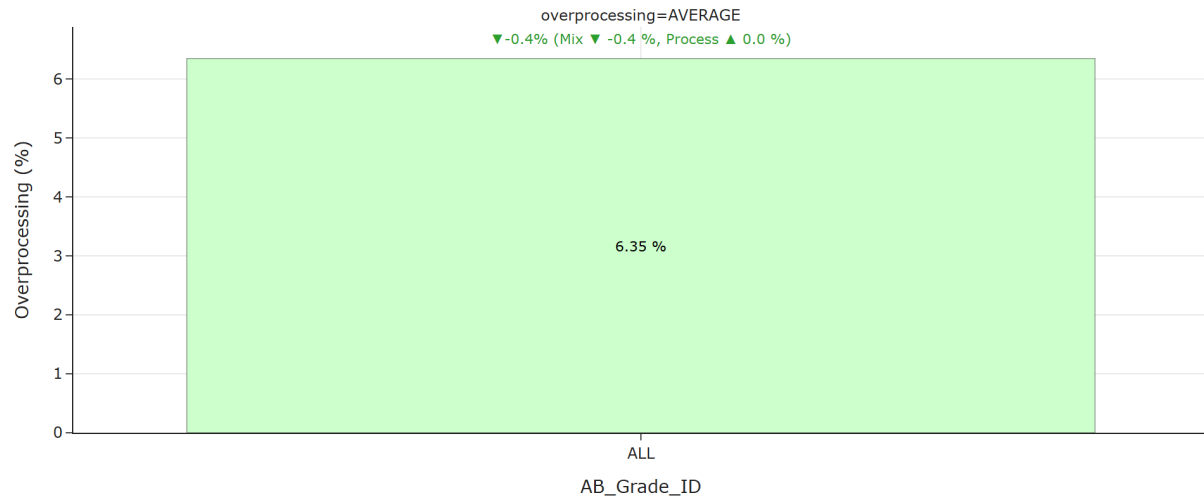
with the biggest drop in grade 3200115 (-0.23 €/t). These components stand out and may require further investigation.



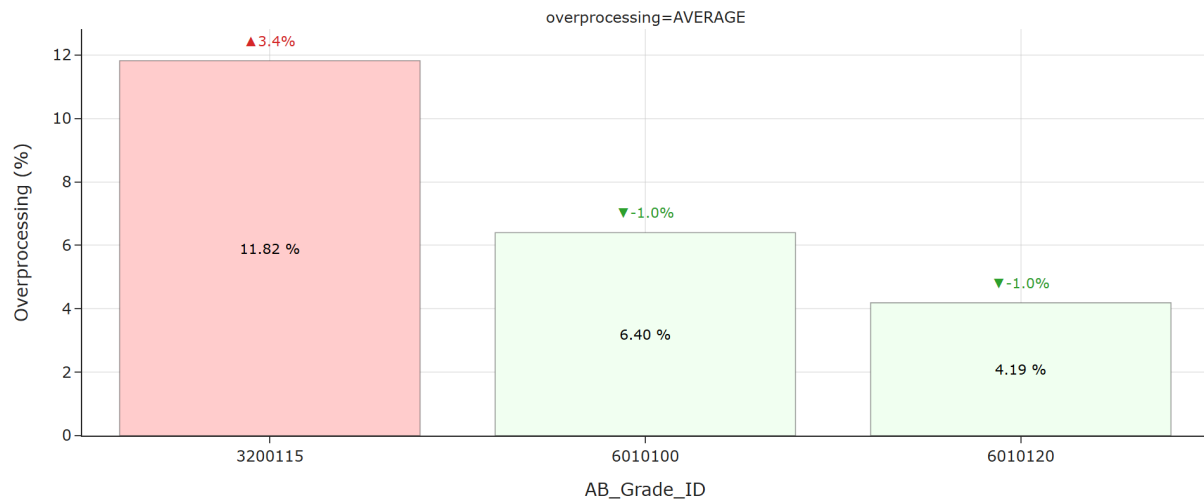
Average change by chemicals:

- **Sizing_Agent_€/T** increased on average (+0.60 €/t).
- **Bentonite_2_€/T** decreased on average (-0.14 €/t).
- **Fixative_2_€/T** decreased on average (-0.13 €/t).
- **Natriumhydroxide_€/T** increased on average (+0.07 €/t).
- **Retention_Aid_€/T** increased on average (+0.06 €/t).
- **Bentonite_1_€/T** increased on average (+0.04 €/t).
- **Defoamer_€/T** increased on average (+0.04 €/t).
- **Deaerator_€/T** decreased on average (-0.02 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Sizing_Agent_€/T** (+0.60 €/t on average), with the highest increase in grade 3200115 (+0.60 €/t) and the largest average decrease comes from **Bentonite_2_€/T** (-0.14 €/t on average), with the biggest drop in grade 3200115 (-0.21 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw a decrease of 0.38 %, resulting in 6.35 %.

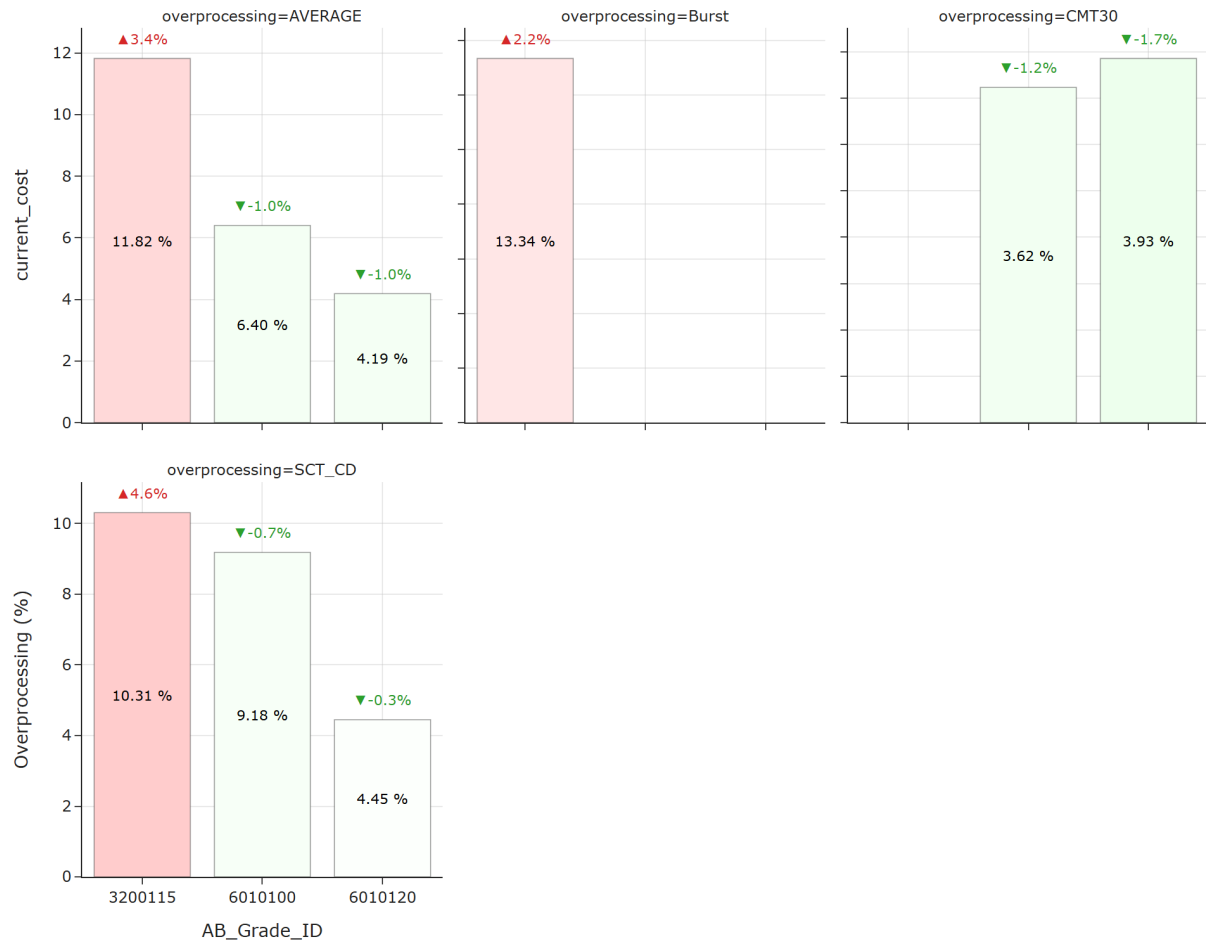


For grade 3200115, TOTAL overprocessing saw an increase of 3.42 %, resulting in 11.82 %.

For grade 6010100, TOTAL overprocessing saw a decrease of 0.96 %, resulting in 6.40 %.

For grade 6010120, TOTAL overprocessing saw a decrease of 0.98 %, resulting in 4.19 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 3200115 (+3.42 %) and the strongest overprocessing decrease was observed for grade 6010120 (-0.98 %). These movements stand out and may require further investigation.



Average change by component:

- **Burst** increased on average (+2.24 %).
- **CMT30** decreased on average (-1.44 %).
- **SCT_CD** increased on average (+1.20 %).

A more detailed breakdown by component shows that the largest average increase comes from **Burst** (+2.24 % on average), with the highest increase in grade 3200115 (+2.24 %) and the largest average decrease comes from **CMT30** (-1.44 % on average), with the biggest drop in grade 6010120 (-1.66 %). These components stand out and may require further investigation.

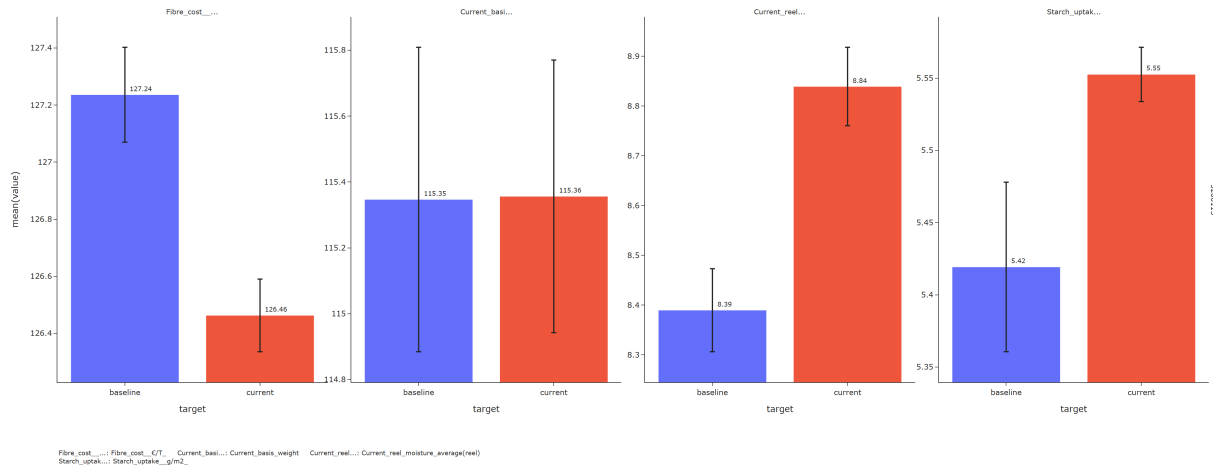
3 Breakdown by grades

3.1 3200115

3.1.1 Fiber cost

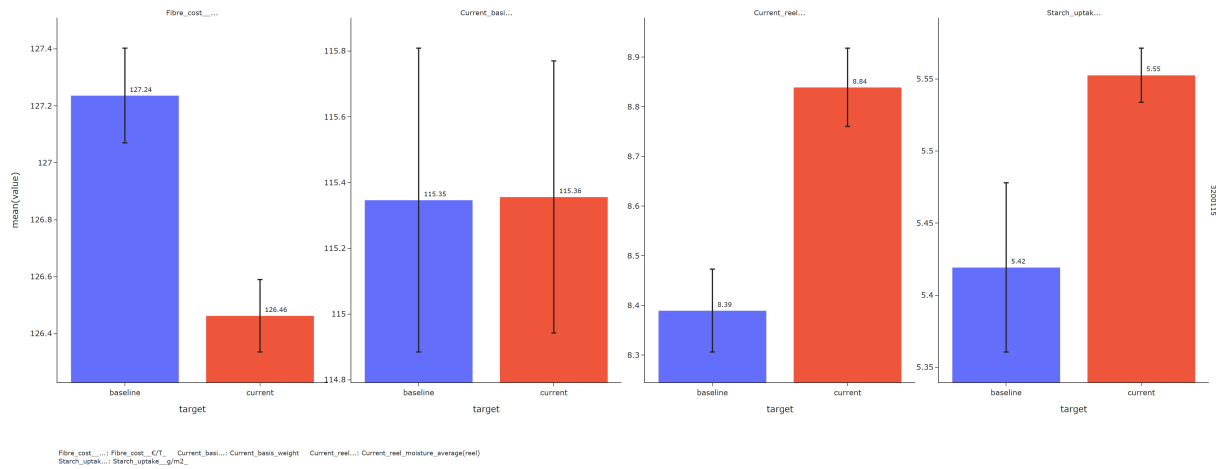


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Current reel moisture average(reel) increased (+0.45)



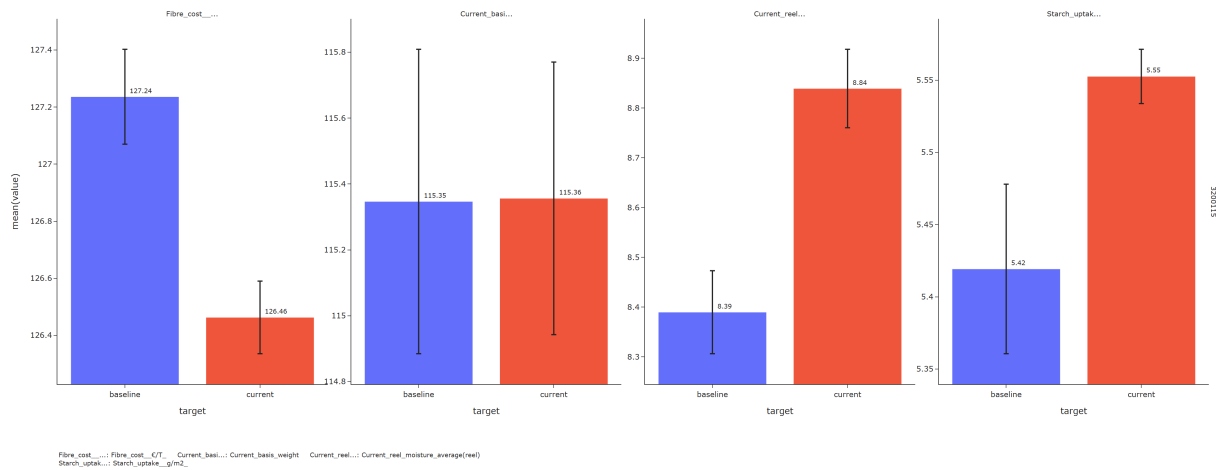
3.1.2 Steam cost

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## KeyError: "['Uhle_box_1_flow___l/min_', 'Uhle_box_2_flow___l/min_'] not in index"
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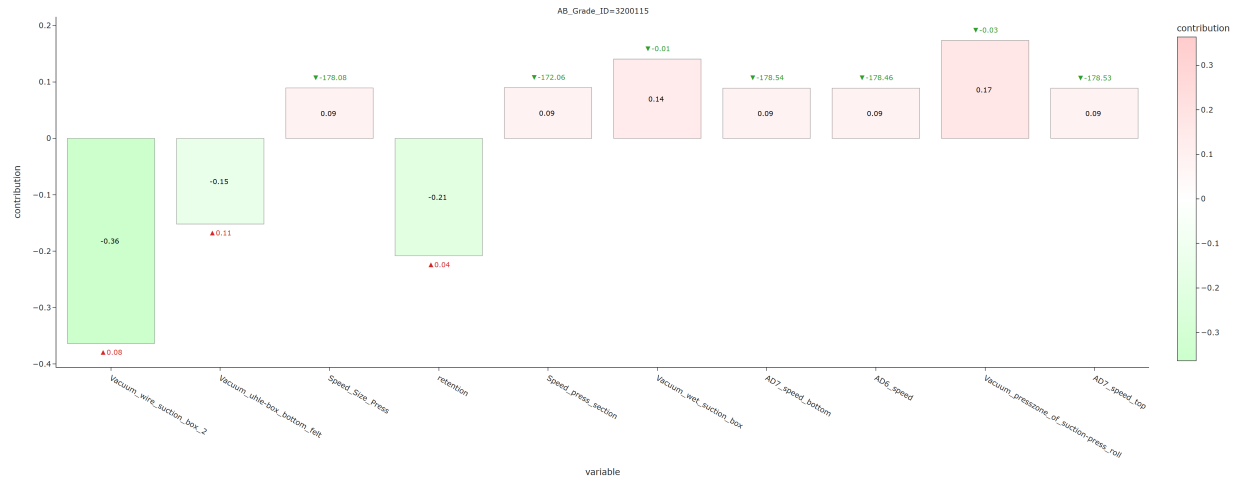


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Current reel moisture average(reel) increased (+0.45)

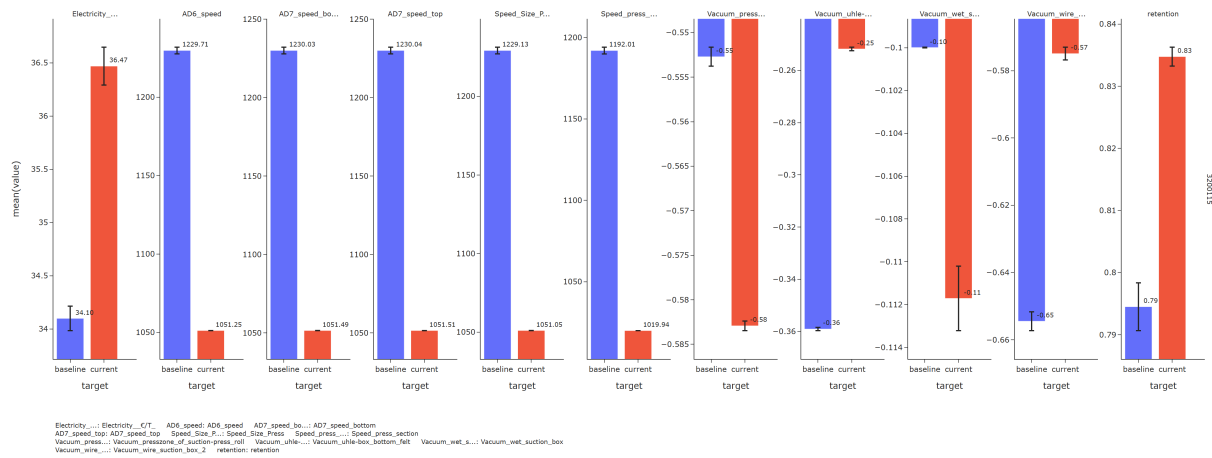
KeyError: "['Steam_€/T_'] not in index"



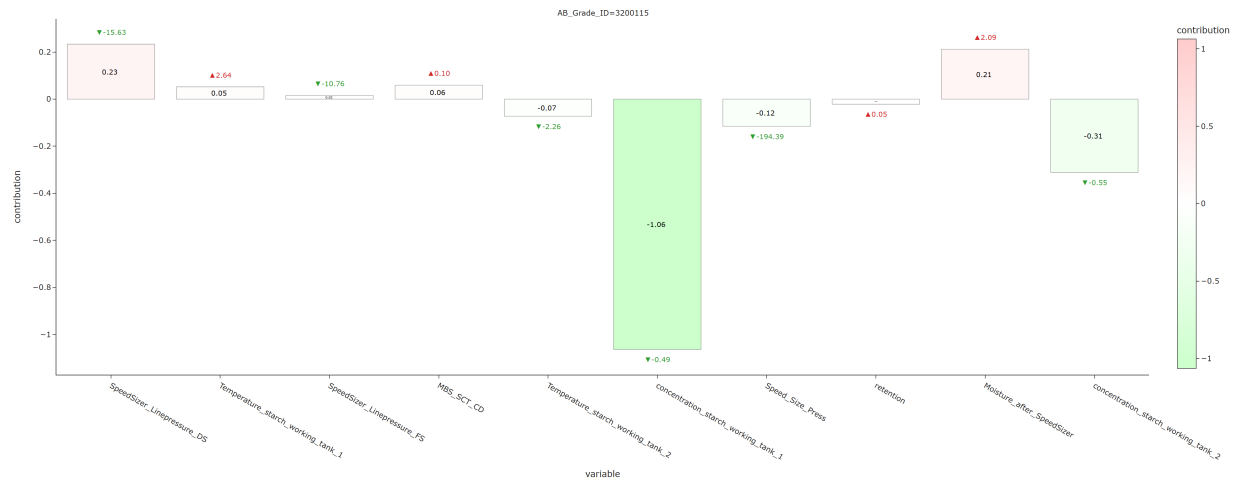
3.1.3 Electricity cost



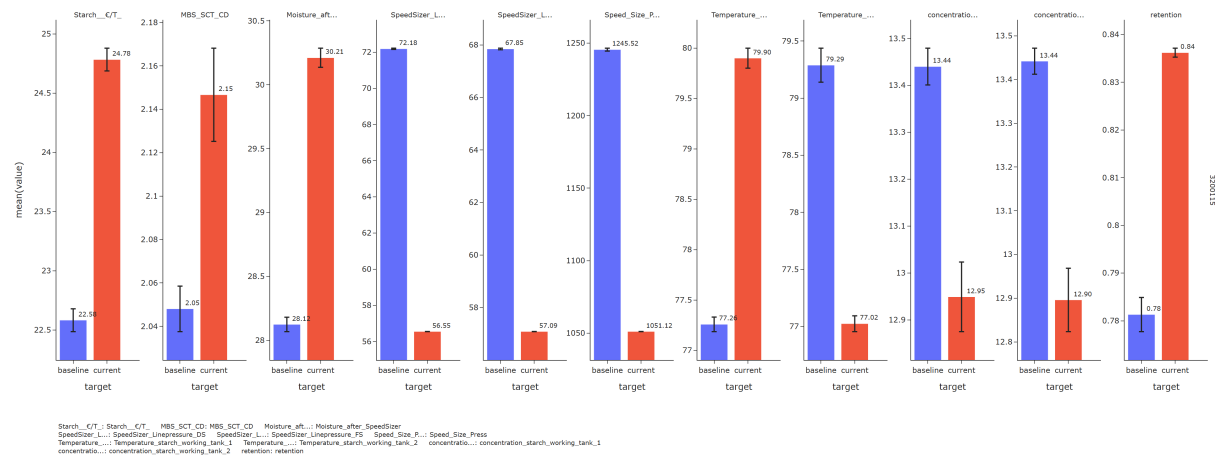
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Vacuum wire suction box 2 increased (+0.08) - Retention increased (+0.04) - Vacuum uhle-box bottom felt increased (+0.11) **Cost-increasing drivers:** - Vacuum presszone of suction-press roll decreased (-0.03) - Vacuum wet suction box decreased (-0.01) - Speed press section decreased (-172.06)



3.1.4 Starch cost



For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Concentration starch working tank 1 decreased (-0.49) - Concentration starch working tank 2 decreased (-0.55) **Cost-increasing drivers:** - SpeedSizer Linepressure DS decreased (-15.63)

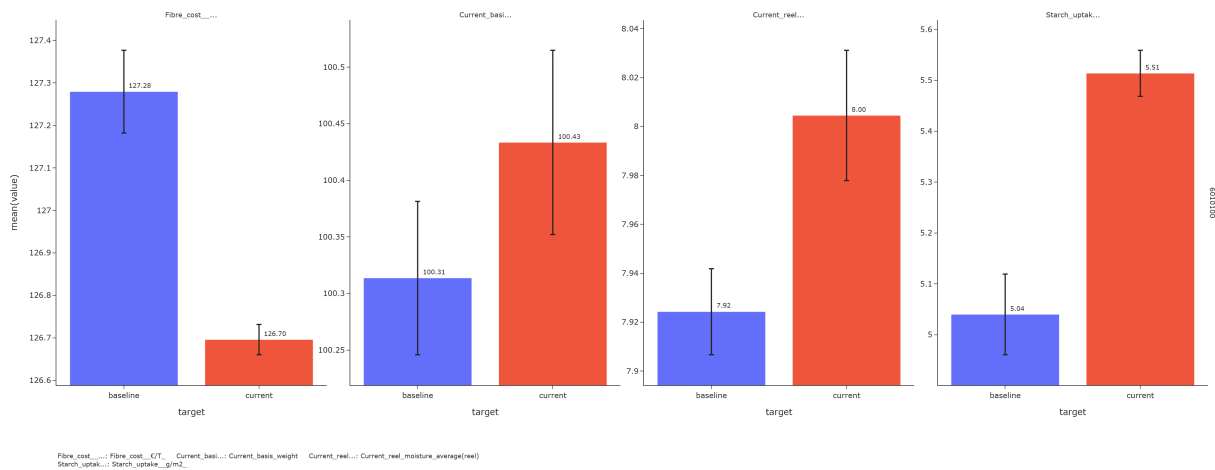


3.2 6010100

3.2.1 Fiber cost

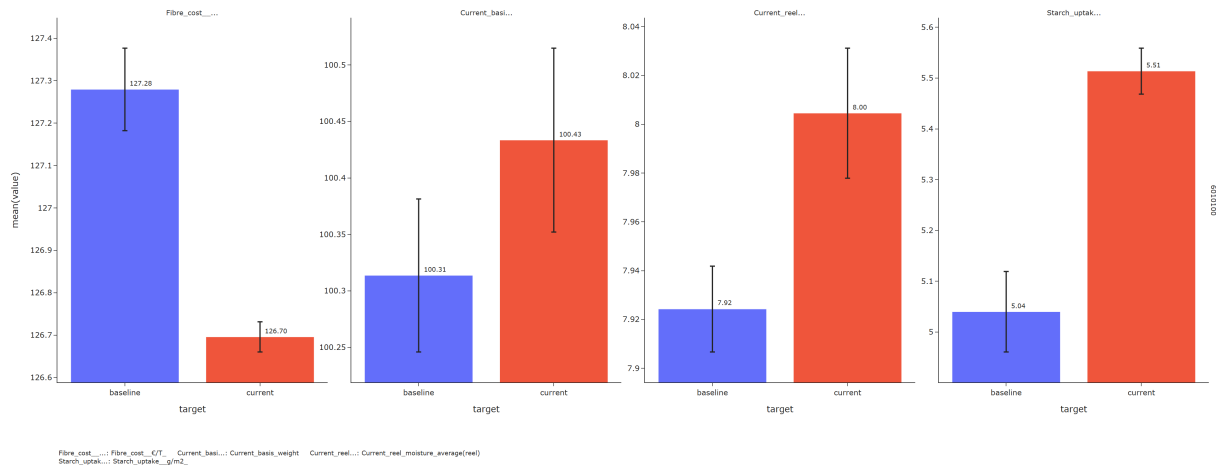


For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Starch uptake g/m2 increased (+0.47)



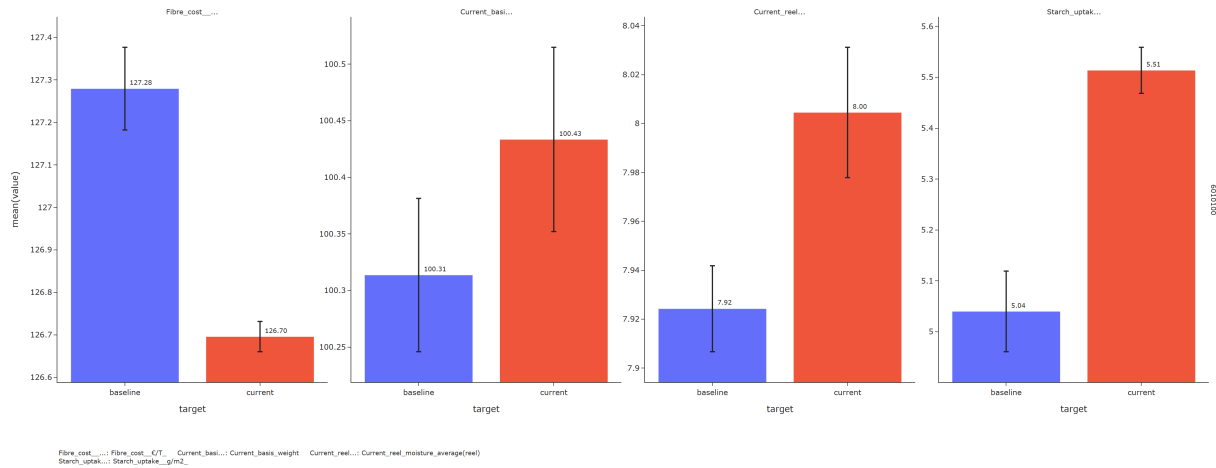
3.2.2 Steam cost

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## KeyError: "['Uhle_box_1_flow__l/min_', 'Uhle_box_2_flow__l/min_'] not in index"
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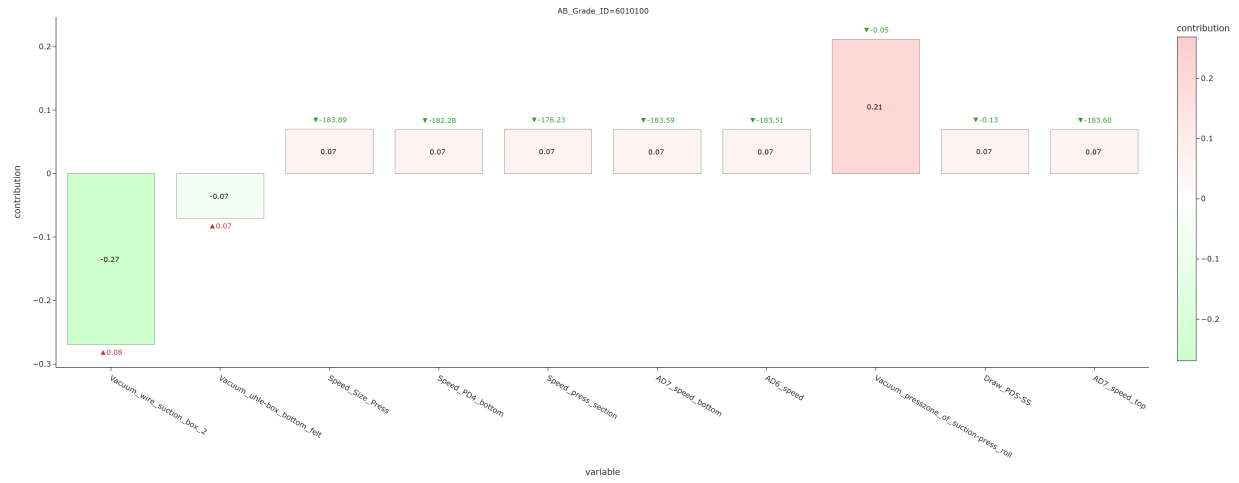


For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Starch uptake g/m2 increased (+0.47)

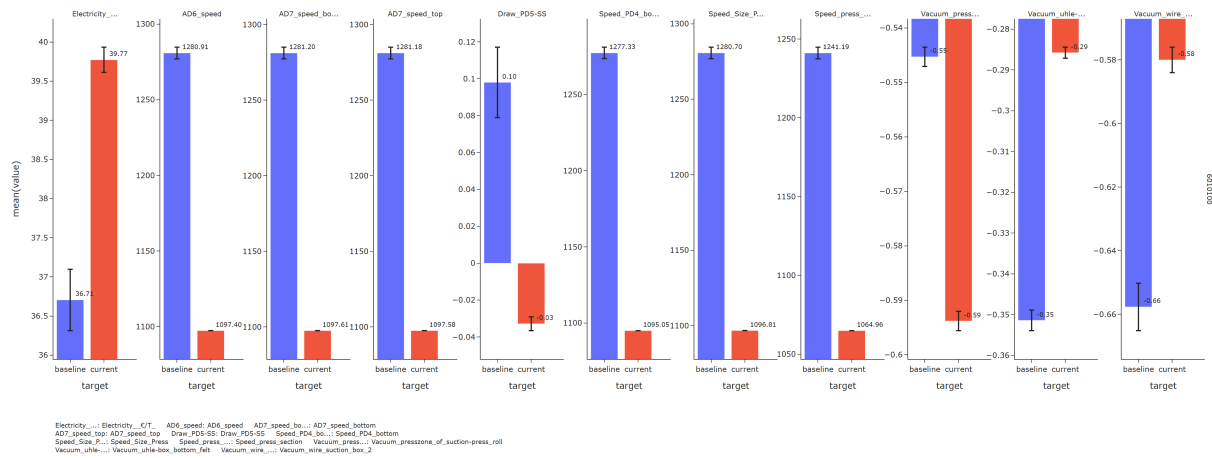
KeyError: "['Steam_€/T_'] not in index"



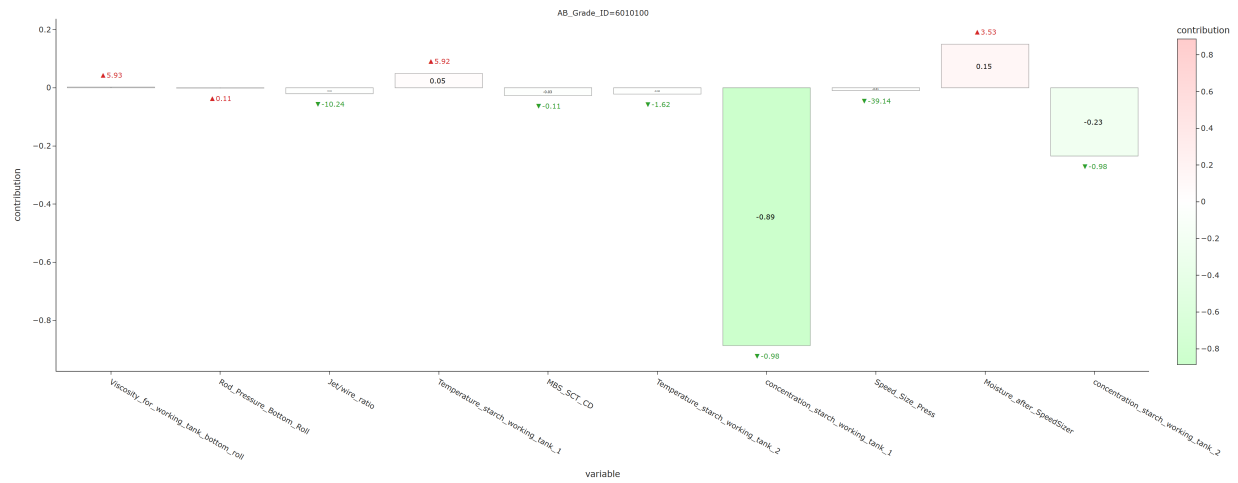
3.2.3 Electricity cost



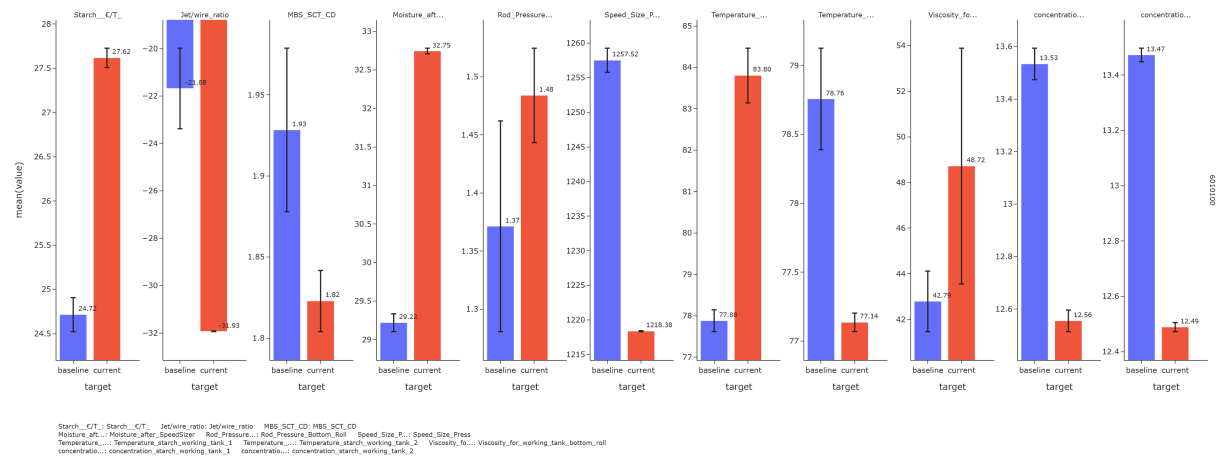
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Vacuum wire suction box 2 increased (+0.08) - Vacuum uhle-box bottom felt increased (+0.07) **Cost-increasing drivers:** - Vacuum presszone of suction-press roll decreased (-0.05) - Speed Size Press decreased (-183.89) - Speed press section decreased (-176.23)



3.2.4 Starch cost

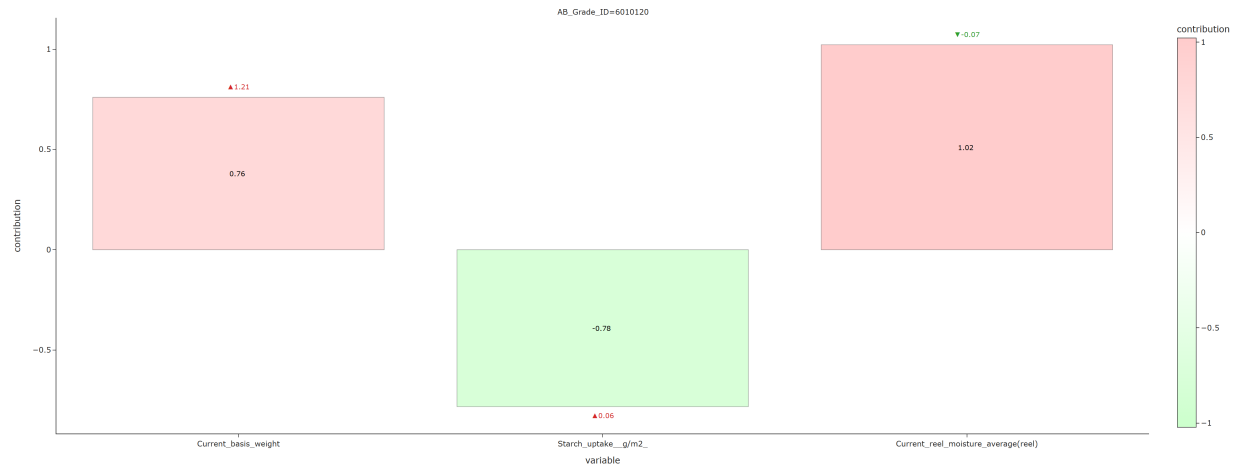


For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Concentration starch working tank 1 decreased (-0.98) - Concentration starch working tank 2 decreased (-0.98) **Cost-increasing drivers:** - Moisture after SpeedSizer increased (+3.53)

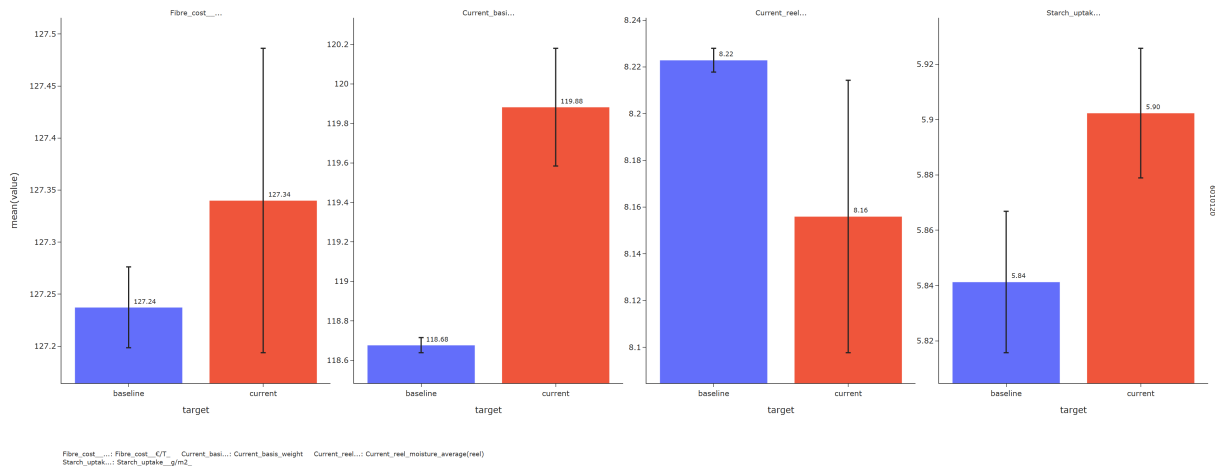


3.3 6010120

3.3.1 Fiber cost



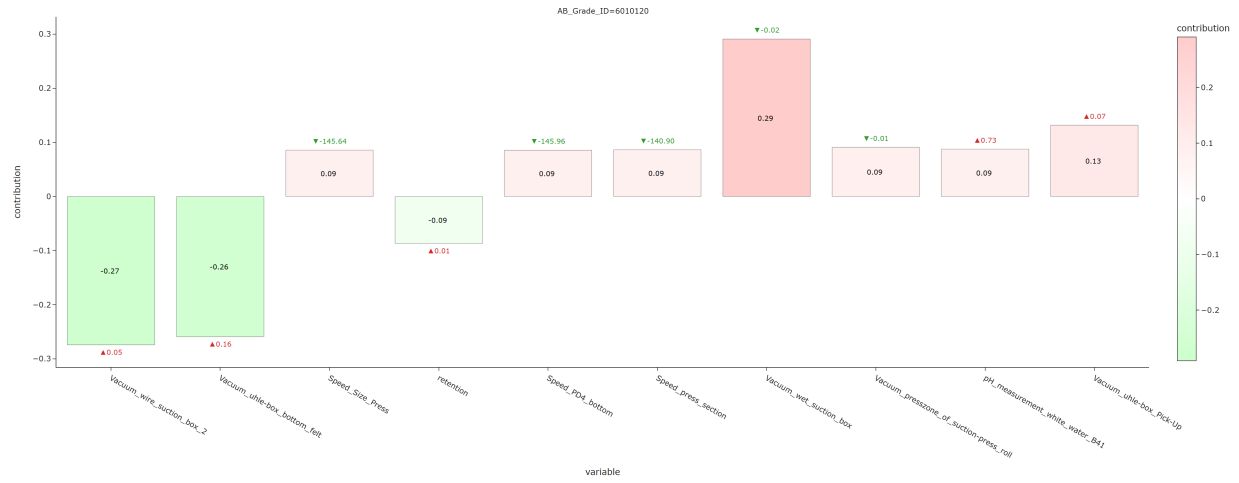
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers:** - Current reel moisture average(reel) decreased (-0.07)



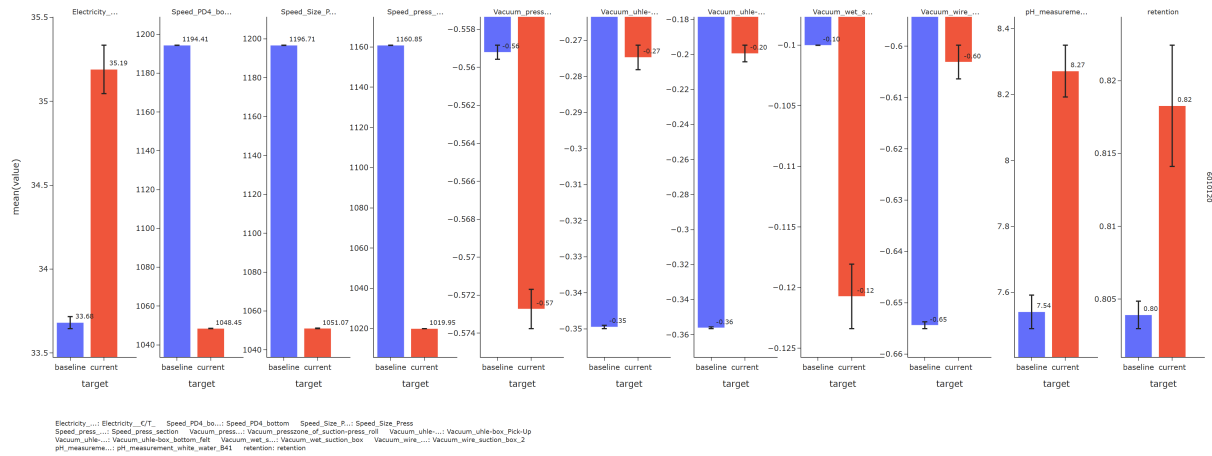
3.3.2 Steam cost

KeyError: "['Uhle_box_1_flow__l/min_', 'Uhle_box_2_flow__l/min_'] not in index"

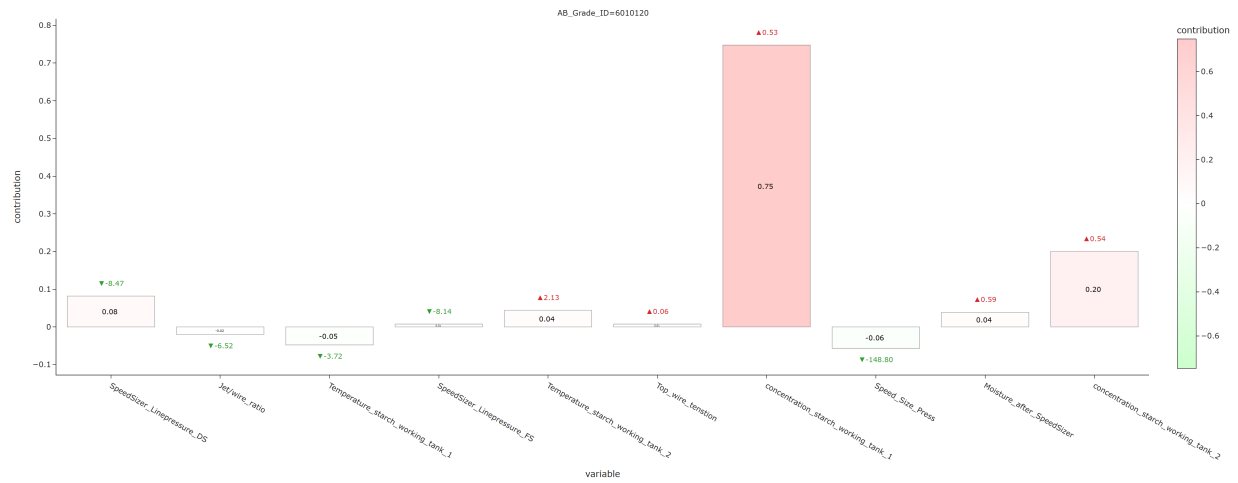
3.3.3 Electricity cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Vacuum wire suction box 2 increased (+0.05) - Vacuum uhle-box bottom felt increased (+0.16) **Cost-increasing drivers:** - Vacuum wet suction box decreased (-0.02) - Vacuum uhle-box Pick-Up increased (+0.07) - Vacuum presszone of suction-press roll decreased (-0.01) - PH measurement white water B41 increased (+0.73)



3.3.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers:** - Concentration starch working tank 1 increased (+0.53) - Concentration starch working tank 2 increased (+0.54) - SpeedSizer Linepressure DS decreased (-8.47)

